

Yulong Zhao

ylzhao@arizona.edu | LinkedIn | Website

Education

University of Arizona — PhD in Statistics	2023 – Expected 05/2028
University of Wisconsin–Madison — MS in Statistics and Data Science	2022 – 2023
University of Science and Technology of China — BS in Mathematics and Applied Mathematics	2018 – 2022

Research Experience

Time-Aware Modeling of User Action Logs Sept 2023 – Present
University of Arizona | Advisor: Prof. Xueying Tang | NSF #2310664

- Developed an **interpretable time-aware model** for process data (timestamped action logs) that tests, for each action, whether its effect on the outcome is absent, constant, or depends on *when* it happens.
- On 19,089 events from 2,668 test-takers in OECD PISA 2012, raised cross-validated accuracy from 93.5% to **96.1%** and **halved log loss** ($0.237 \rightarrow 0.118$) vs. a count-feature baseline, winning on all 10 folds.
- Showed the gain comes from actions whose effect **flips sign with timing** (early exploration vs. final choice), a signal that count features cannot capture.

Latent Skill and Hidden-State Modeling of User Behavior 2026 – Present
University of Arizona | Advisor: Prof. Xueying Tang

- Designed a **structured latent-variable framework** that separates stable user skills from evolving problem-solving states, generating each step as hidden state $\rightarrow$ interface choice $\rightarrow$ action (a skill-driven **hidden Markov model**).
- Unlike prior models that fix states or skill-behavior links in advance, the framework **learns from data alone** which skills drive which actions, interface choices, and state transitions, through sparse skill-to-behavior maps.
- On 1,340 users of an email-and-spreadsheet task, it discovered skills, states, and skill-behavior links that are **substantively interpretable** without any manual labeling.

Real-Time Early Warning with a Controlled False-Alarm Rate 2026 – Present
University of Arizona | Advisor: Prof. Xueying Tang

- Built a **real-time risk-scoring** system that turns each user's partial action sequence in OECD PISA problem-solving tasks into an evolving probability of success, flagging users heading toward failure while they are still working.
- Developed **concurrently** with a 2026 **JASA** paper (Yuan & Fang) built on the same core idea, machine-learned success-probability curves; this project goes further by deciding *when* to alert, capping false alarms at $\leq 5\%$.
- Compared GRU, temporal CNN, and Transformer models with leakage-free logistic baselines; the best temporal CNN caught **98.9%** of failing users at precision ≥ 0.98 , with false alarms $\leq 4\%$ on all 3 tasks, vs. 85.6% for GRU.

Time-Series Forecasting and Regime-Shift Detection May 2024 – Present
University of Arizona | IMSI Interdisciplinary Research Cluster | NSF #1929348

- Built a **walk-forward forecasting** pipeline (RBF-SVR, time-ordered CV, no future-information leakage) on PCA-compressed features from 8 environmental series; beat the historical-mean baseline in **18/18** settings with 38% lower MSE.
- Developed a **regime-shift detector** based on **recurrence plots** of multivariate environmental series and identified statistically significant shifts after multiple-testing correction, the strongest aligning with the **end-Triassic mass extinction** (~ 201 Ma).
- Ran robustness checks that separated stable shifts from a fragile one, and delivered visual reports to geoscience collaborators.

Publications and Presentations

Zhao, Y., & Tang, X. Scalar-on-process model for educational assessment: modeling time-varying action effects with group LASSO regularization. *Manuscript ready for submission.*

Refereed oral presentation of the above, Symposium on Data Science & Statistics (SDSS), Milwaukee, WI, April 2026.

Skills

Programming: Python (NumPy, pandas, scikit-learn, Numba), R (Keras/TensorFlow), SQL, Git, Slurm/HPC, LaTeX.

Machine learning: sequence models (GRU, TCN, Transformer), regularized GLMs, kernel methods, latent-variable models/EM.

Statistics: threshold calibration, leakage-safe validation, bootstrap, multiple testing, simulation studies.

Awards

Research Fellowship (2024–26) & Incoming PhD Award, U. Arizona; Academic Excellence Award, UW–Madison; Outstanding Undergraduate Award, USTC.